

computing power [Nordhaus 2001], engineers have become vastly more proficient at applying the loom's basic capabilities—rapid execution of stored instructions—to a panoply of tasks. How does this advance affect the task composition of human work? The answer depends both upon how computers substitute for or complement workers in carrying out specific tasks, and how these tasks substitute for one another. We illustrate these cases by considering the application of computers to routine and nonroutine cognitive and manual tasks.

In our usage, a task is “routine” if it can be accomplished by machines following explicit programmed rules. Many manual tasks that workers used to perform, such as monitoring the temperature of a steel finishing line or moving a windshield into place on an assembly line, fit this description. Because these tasks require methodical repetition of an unwavering procedure, they can be exhaustively specified with programmed instructions and performed by machines.

A problem that arises with many commonplace manual and cognitive tasks, however, is that the procedures for accomplishing them are not well understood. As Polanyi [1966] observed, “We can know more than we can tell [p. 4] . . . The skill of a driver cannot be replaced by a thorough schooling in the theory of the motorcar; the knowledge I have of my own body differs altogether from the knowledge of its physiology; and the rules of rhyming and prosody do not tell me what a poem told me, without any knowledge of its rules [p. 20].” We refer to tasks fitting Polanyi's description as “nonroutine,” that is, tasks for which the rules are not sufficiently well understood to be specified in computer code and executed by machines. Navigating a car through city traffic or deciphering the scrawled handwriting on a personal check—minor undertakings for most adults—are not routine tasks by our definition (see Beamish, Levy, and Murnane [1999] and Autor, Levy and Murnane [2002] for examples). The reason is that these tasks require visual and motor processing capabilities that cannot at present be described in terms of a set of programmable rules [Pinker 1997].⁴

Our conceptual model suggests that, because of its declining cost, computer-controlled machinery should have substantially

4. If a manual task is performed in a well-controlled environment, however, it can often be automated despite the seeming need for adaptive visual or manual processing. As Simon [1960] observed, environmental control is a substitute for flexibility.

substituted for workers in performing routine manual tasks. This phenomenon is not novel. Substitution of machinery for repetitive human labor has been a thrust of technological change throughout the Industrial Revolution [Hounshell 1985; Mokyr 1990; Goldin and Katz 1998]. By increasing the feasibility of machine substitution for repetitive human tasks, computerization furthers—and perhaps accelerates—this long-prevailing trend.

The advent of computerization also marks a qualitative enlargement in the set of tasks that machines can perform. Because computers can perform symbolic processing—storing, retrieving, and acting upon information—they augment or supplant human cognition in a large set of information-processing tasks that historically were not amenable to mechanization. Over the last three decades, computers have substituted for the calculating, coordinating, and communicating functions of bookkeepers, cashiers, telephone operators, and other handlers of repetitive information-processing tasks [Bresnahan 1999].

This substitution marks an important reversal. Previous generations of high technology capital sharply increased demand for human input of routine information-processing tasks, as seen in the rapid rise of the clerking occupation in the nineteenth century [Chandler 1977; Goldin and Katz 1995]. Like these technologies, computerization augments demand for clerical and information-processing tasks. But in contrast to its nineteenth century predecessors, it permits these tasks to be automated.

The capability of computers to substitute for workers in carrying out cognitive tasks is limited, however. Tasks demanding flexibility, creativity, generalized problem-solving, and complex communications—what we call nonroutine cognitive tasks—do not (yet) lend themselves to computerization [Bresnahan 1999]. At present, the need for explicit programmed instructions appears a binding constraint. There are very few computer-based technologies that can draw inferences from models, solve novel problems, or form persuasive arguments.⁵ In the words of computer scientist Patrick Winston

5. It is, however, a fallacy to assume that a computer must reproduce all of the functions of a human to perform a task traditionally done by workers. For example, Automatic Teller Machines have supplanted many bank teller functions, although they cannot verify signatures or make polite conversation while tallying change. Closely related, computer capital may substitute for the routine components of predominantly nonroutine tasks, e.g., on-board computers that direct taxi cabs. What is required for our conceptual model is that the routine and nonroutine tasks embodied in a job are imperfect substitutes. Consequently, a decline in the price of accomplishing routine tasks does not eliminate demand for nonroutine tasks.

[1999]: "The goal of understanding intelligence, from a computational point of view, remains elusive. Reasoning programs still exhibit little or no common sense. Today's language programs translate simple sentences into database queries, but those language programs are derailed by idioms, metaphors, convoluted syntax, or ungrammatical expressions."⁶

The implication of our discussion is that because present computer technology is more substitutable for workers in carrying out routine tasks than nonroutine tasks, it is a relative complement to workers in carrying out nonroutine tasks. From a production function standpoint, outward shifts in the supply of routine informational inputs, both in quantity and quality, increase the marginal productivity of workers performing nonroutine tasks that demand these inputs. For example, comprehensive bibliographic searches increase the quality of legal research and timely market information improves the efficiency of managerial decision-making. More tangibly, because repetitive, predictable tasks are readily automated, computerization of the workplace raises demand for problem-solving and communications tasks such as responding to discrepancies, improving production processes, and coordinating and managing the activities of others. This changing allocation of tasks was anticipated by Drucker [1954] in the 1950s: "The technological changes now occurring will carry [the Industrial Revolution] a big step further. They will not make human labor superfluous. On the contrary, they will require tremendous numbers of highly skilled and highly trained men—managers to think through and plan, highly trained technicians and workers to design the new tools, to produce them, to maintain them, to direct them" [p. 22, brackets added].

Table I provides examples of tasks in each cell of our two-by-two matrix of workplace tasks (routine versus nonroutine, manual versus information processing) and states our hypothesis about the impact of computerization for each cell. The next section formalizes these ideas and derives empirical implications.⁷

6. Software that recognizes patterns (e.g., neural networks) or solves problems based upon inductive reasoning from well-specified models is under development. But these technologies have had little role in the "computer revolution" of the last several decades. As one example, current speech recognition software based on pattern recognition can recognize words and short phrases but can only process rudimentary conversational speech [Zue and Glass 2000].

7. Our focus on task shifts in the process of production within given jobs overlooks two other potentially complementary avenues by which technical

TABLE I
PREDICTIONS OF TASK MODEL FOR THE IMPACT OF COMPUTERIZATION ON FOUR
CATEGORIES OF WORKPLACE TASKS

	Routine tasks	Nonroutine tasks
	Analytic and interactive tasks	
Examples	• Record-keeping • Calculation • Repetitive customer service (e.g., bank teller)	• Forming/testing hypotheses • Medical diagnosis • Legal writing • Persuading/selling • Managing others
Computer impact	• Substantial substitution	• Strong complementarities
	Manual tasks	
Examples	• Picking or sorting • Repetitive assembly	• Janitorial services • Truck driving
Computer impact	• Substantial substitution	• Limited opportunities for substitution or complementarity

I.A. *The Demand for Routine and Nonroutine Tasks*

The informal task framework above implies three postulates about how computer capital interacts with human labor input.

- A1. Computer capital is more substitutable for human labor in carrying out routine tasks than nonroutine tasks.
- A2. Routine and nonroutine tasks are themselves imperfect substitutes.
- A3. Greater intensity of routine inputs increases the marginal productivity of nonroutine inputs.

To develop the formal implications of these assumptions, we write a simple, general equilibrium production model with two

change impacts job task demands. First, innovations in the organization of production reinforce the task-level shifts that we describe above. See Adler [1986], Zuboff [1988], Levy and Murnane [1996], Acemoglu [1999], Bresnahan [1999], Bartel, Ichniowski, and Shaw [2000], Brynjolfsson and Hitt [2000], Lindbeck and Snower [2000], Mobius [2000], Thesmar and Thoenig [2000], Caroli and Van Reenen [2001], Fernandez [2001], Autor, Levy, and Murnane [2002], and Bresnahan, Brynjolfsson, and Hitt [2002] for examples. Second, distinct from our focus on process innovations, Xiang [2002] presents evidence that product innovations over the past 25 years have also raised skill demands.